

The three- ρ piece of $|\mathfrak{B}_2|^2$

Structure check — where the transverse divergence sits — the p^+ integral

$$A \equiv (x - z)^2, \quad B \equiv (x - w)^2, \quad C \equiv (z - w)^2, \quad S \equiv p^+ + k^+, \quad D \equiv p^+ A + k^+ B, \quad P_\vee \equiv \vee - k^+$$

Verdict. It is correct. Both prefactors reproduce exactly, including the relative minus sign, which is what makes this piece **real on its own** — no "+ c.c." is needed (§1). The colour is right in both brackets, and you were right *not* to pull f out: the two ket terms carry f^{cbd} and f^{acd} , genuinely different tensors (§2). One transpose to fix. On divergences: **no ultraviolet divergence anywhere** — all 57 collapsing regions converge, before *and* after the p^+ integration. The single transverse divergence sits at $|z| \rightarrow \infty$, it is **infrared**, and its coefficient is known in closed form and confirmed to 8 decimals (§3). The p^+ integral is clean — no on-contour pole, unlike the $\mathfrak{B}_3$ row — and reduces to three elementary masters (§4).

1. Prefactors, and why this piece is real

Split $\mathfrak{B}_2 = \mathfrak{B}^{(1)} + \mathfrak{B}^{(2)}$ with $\mathfrak{B}^{(1)} : \frac{ig^2 \sqrt{p^+ k^+}}{4\pi^2 S}$ (one ρ , **imaginary**) and $\mathfrak{B}^{(2)} : -\frac{g^2}{8\pi^2 \sqrt{p^+ k^+}}$ (two ρ , **real**). The three- ρ piece is the sum of the two cross terms:

your term	which cross term	computed coefficient	vs quoted
first (−)	$\mathfrak{B}^{(2)\dagger} \mathfrak{B}^{(1)}$	$-\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^4} \frac{1}{S}$	1
second (+)	$\mathfrak{B}^{(1)\dagger} \mathfrak{B}^{(2)}$	$+\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^4} \frac{1}{S}$	1

Because $\mathfrak{B}^{(1)}$ is imaginary and $\mathfrak{B}^{(2)}$ real, the two cross terms are complex conjugates of one another and must carry **opposite signs**. Your $-\dots +$ is exactly right, and it means

$$\frac{d^3 N_1}{d^3 k} = 2i \operatorname{Im}(\dots)$$

is **real by itself**. (Contrast the $\mathfrak{A}^{(3)}$ row, where only one ordering was written and a + c.c. was required.)

The $\sqrt{p^+ k^+}$ and $1/\sqrt{p^+ k^+}$ cancel, so no $p^+ k^+$ survives, and the vertex's overall $\frac{1}{S}$ distributes correctly:

$$\frac{1}{S} \left[\frac{\delta^{ij}(A-B)}{2C} + \frac{S}{k^+}(\dots) + \frac{S}{p^+}(\dots) \right] = \frac{\delta^{ij}(A-B)}{2SC} + \frac{1}{k^+}(\dots) + \frac{1}{p^+}(\dots) \quad \checkmark$$

2. Colour

$\mathfrak{B}_2^{bc}$'s one- ρ piece carries $f^{cba} \rho^a(x)$ (free b, c). Relabel $a \rightarrow d$:

$$U^{ac}(w) \mathfrak{B}^{bc(1)} \rightarrow f^{cbd} U^{ac}(w) \rho^d(x), \quad -U^{\dagger bc}(z) \bar{\mathfrak{B}}^{ca(1)} \rightarrow -f^{acd} U^{bc}(z) U^{de}(x) \rho^e(x).$$

f is not an overall factor here, and you correctly left it inside. The two terms carry f^{cbd} and f^{acd} — different tensors ($\|f^{abc} - f^{bac}\| = 9.798$). This is the opposite situation from the $\mathfrak{B}_3$ row, where f *could* be pulled out, and from the $\mathfrak{A}^{(3)}$ row, where it was pulled out but should not have been. All three are handled correctly in what you have written here.

The two- ρ piece carries $\{\rho^c(y), \rho^b(x)\}$; with the relabelling $y \rightarrow x'$ (the i -slot, paired with w') and $x \rightarrow y$ (the j -slot, paired with z) this becomes $\{\rho^{c'}(x'), \rho^b(y)\}$ — your bracket. $\checkmark$

One transpose. The row has $U^{c'a}(w')$; you wrote $U^{ac'}(w')$. Same as in the last three rows — pick one ordering and use it in both terms of the bracket, otherwise the difference is not a well-defined matrix difference.

The two cross terms are each other's mirror, as they should be: in the first the one- ρ vertex is in the ket (structures $(x - w)$, $(x - z)$, $(z - w)$, denominator D) and the two- ρ in the bra; in the second they swap, giving $(x - w')$, $(z - w')$ and kernel $\frac{(x' - w)^i (y - z)^j}{(x' - w)^2 (y - z)^2}$. Both as written. $\checkmark$

3. Where the transverse divergence is

3.1 Not in the ultraviolet

Every subset of $\{x, y, z, w, x', w'\}$ of size ≥ 2 collapsed at scale ρ against the measure $\rho^{2(n-1)} \frac{d\rho}{\rho}$ — **all 57 regions converge**. Checked twice: on the integrand as written, *and* on the p^+ -integrated form of §4, where the logarithms $\log \frac{k^+ B}{\Lambda A}$ and $\log \frac{\sqrt{A}}{k^+ B}$ could in principle have created new short-distance behaviour. They do not: near $x \rightarrow w$ or $x \rightarrow z$ the logarithm goes as $\log \rho$ while the accompanying tensor supplies at most $\frac{1}{\rho}$, so the integrand is $\frac{\log \rho}{\rho}$ against $\rho d\rho$.

3.2 It is at $|z| \rightarrow \infty$

variable	$R = 10^2$	10^4	10^6	status
$ x $	0.6363502	0.6364102	0.6364100	log divergent
$ y $			noise	convergent
$ z $	1.2353	1.2355032	1.2355032	log divergent
$ w $	1.4490730	1.4490410	1.4490410	log divergent (bare)
$ x' , w' $			noise	convergent

Sorting them:

- $|w|$ (and $|w'|$) carry the phase $e^{-ik(w'-w)}$, which gives the Bessel $R^{-1/2}$ envelope — convergent once the phase is restored.
- $|x|$ is a source position; the integral is cut off by the support of $\rho(x)$.
- $|z|$ **is the one that is not protected**. It is a genuine integration variable, it carries no phase, and no Wilson line damps it.

3.3 The mechanism and the exact coefficient

At large $|z|$ the one- ρ vertex degenerates — the $\frac{S}{k^+}$ structure grows like $|z|$ while $D \sim p^+ |z|^2$ — and the whole bracket collapses onto a product of two Weizsäcker–Williams kernels:

$$\frac{1}{D} \left[\frac{\delta^{ij}(A-B)}{2SC} + \frac{\mathcal{P}^{ij}}{p^+} + \frac{\mathcal{K}^{ij}}{k^+} \right] \xrightarrow{|z| \rightarrow \infty} -\frac{1}{2p^+ k^+} \frac{(x-w)^i}{(x-w)^2} \frac{(x-z)^j}{(x-z)^2}.$$

Then $\frac{(y-z)^j}{(y-z)^2} \rightarrow -\frac{\hat{z}^j}{|z|}$ and $\frac{(x-z)^j}{(x-z)^2} \rightarrow -\frac{\hat{z}^j}{|z|}$, so the j -contraction gives $\frac{1}{|z|^2}$ with **no angular cancellation** — hence the logarithm. Its coefficient:

$\lim_{ z \rightarrow \infty} \langle \text{integrand} \rangle_\theta z ^2 = -\frac{1}{2p^+ k^+} \frac{(x'-w') \cdot (x-w)}{(x'-w')^2 (x-w)^2}$						
	$ z $	10^2	10^3	10^4	10^5	10^6
measured		-1.23527	-1.2355020	-1.2355032	-1.2355032	-1.2355032
						-1.23550324

What to do with it. It is **infrared**, not ultraviolet — there is no short-distance counterterm for it. Regulate it as in the previous rows, $\int d^{2-2\epsilon} z$ with $\epsilon > 0$, giving a $\frac{1}{\epsilon_{\text{IR}}}$ pole. But note *what* the coefficient is: a product of two WW kernels, $\frac{(x'-w') \cdot (x-w)}{(x'-w')^2 (x-w)^2}$, carrying $\frac{1}{p^+ k^+}$ — precisely the structure of the **four- ρ term** $(2\rho) \times (2\rho)$ that is absent from your decomposition. The tails of the three pieces are built from the same object, so restore the 4ρ term and combine before regulating.

4. The p^+ integral

All the p^+ dependence is $\frac{1}{D}$ times the three bracket weights $\frac{1}{2S}, \frac{1}{p^+}, \frac{1}{k^+}$, since

$$\mathcal{P}^{ij} = \frac{(x-w)^i (z-w)^j}{C} + \frac{(x-w)^i (x-z)^j}{2A}, \quad \mathcal{K}^{ij} = \frac{(z-w)^i (x-z)^j}{C} - \frac{(x-w)^i (x-z)^j}{2B}$$

are both p^+ -independent. Note $D = p^+ A + k^+ B$ has **both signs positive**, so there is no zero on the contour — this row integrates cleanly, unlike the $\mathfrak{B}_3$ row. Three masters, verified against quadrature at $A = 0.3592$, $B = 2.0885$, $\Lambda = 10^{-8}$, $\vee = 10^7$:

$$\begin{aligned}\mathcal{J}_1 &= \int_{\Lambda}^{P_{\vee}} \frac{dp^+}{p^+ D} = \frac{1}{k^+ B} \log \frac{P_{\vee} k^+ B}{\Lambda (P_{\vee} A + k^+ B)} \xrightarrow{\vee \rightarrow \infty} \frac{1}{k^+ B} \log \frac{k^+ B}{\Lambda A}, \\ \mathcal{J}_2 &= \int_{\Lambda}^{P_{\vee}} \frac{dp^+}{D} = \frac{1}{A} \log \frac{P_{\vee} A + k^+ B}{\Lambda A + k^+ B} \rightarrow \frac{1}{A} \log \frac{\vee A}{k^+ B}, \\ \mathcal{J}_3 &= \int_{\Lambda}^{P_{\vee}} \frac{dp^+}{(p^+ + k^+) D} = \frac{1}{k^+ (B - A)} \log \frac{\vee B}{P_{\vee} A + k^+ B} \rightarrow \frac{1}{k^+ (B - A)} \log \frac{B}{A}.\end{aligned}$$

J1: quad	9.662917027	closed	9.662917027	diff	8e-10
J2: quad	39.971531412	closed	39.971531412	diff	4e-14
J3: quad	1.017938795	closed	1.017938800	diff	5e-09

The δ^{ij} term collapses exactly. Its weight is $\frac{A-B}{2C} \mathcal{J}_3$, and the $(A - B)$ cancels the $(B - A)$:

$$\frac{A - B}{2C} \cdot \frac{1}{k^+ (B - A)} \log \frac{\vee B}{P_{\vee} A + k^+ B} = -\frac{1}{2k^+ C} \log \frac{\vee B}{P_{\vee} A + k^+ B} \rightarrow -\frac{\delta^{ij}}{2k^+ (z - w)^2} \log \frac{(x - w)^2}{(x - z)^2}.$$

Exact to 12 digits, and free of *both* Λ and $\vee$.

4.1 The result

$$\begin{aligned}\frac{d^3 N_1}{d^3 k} &= -\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^4} \frac{1}{2\pi} \int_{x,y,z,w,x',w'} e^{-ik(w'-w)} \frac{(x' - w')^i (y - z)^j}{(x' - w')^2 (y - z)^2} \\ &\times \left\{ -\frac{\delta^{ij}}{2k^+ C} \log \frac{\vee B}{P_{\vee} A + k^+ B} + \mathcal{P}^{ij} \mathcal{J}_1 + \frac{\mathcal{K}^{ij}}{k^+} \mathcal{J}_2 \right\} \times [\text{bra colour}] \times [\text{ket colour}]\end{aligned}$$

plus the mirror term with $w \leftrightarrow w'$, $x \leftrightarrow x'$ and the opposite sign — together, twice the imaginary part, hence real.

4.2 Which logarithm sits where

structure	master	$\log \frac{1}{\Lambda}$ residue	$\log \vee$ residue
$\frac{1}{p^+} \mathcal{P}^{ij}$	$\mathcal{J}_1$	$\frac{1}{k^+ (x - w)^2}$	—
$\frac{1}{k^+} \mathcal{K}^{ij}$	$\mathcal{J}_2$	—	$\frac{1}{(x - z)^2}$
$\frac{\delta^{ij} (A - B)}{2S C}$	$\mathcal{J}_3$	—	—

So the rapidity logarithm is isolated in a single structure, $\mathcal{P}^{ij} \mathcal{J}_1$, and the $[\frac{1}{p^+}]_+$ prescription applies to that one alone. The $\log \vee$ in $\mathcal{J}_2$ is a genuine large- p^+ logarithm, not a rapidity one — it will not be absorbed by evolution and should be checked against the kinematic bound.

zint/src/row12_check.py (§1–2), row12_div.py (§3.1–3.2), row12_final.py (§3.3, §4) — every number above.